Module 2: Agentic AI

Architectures and Design Patterns

Demo - 2

Demo-2: Implementing different agentic AI design patterns

Use Case: AI-Powered Clinical Assistant for Emergency Triage & Decision Support

This demo aims to explore and understand how Agentic AI Design Patterns may be included into an AI-powered healthcare assistant so it may assess symptoms, handle patient questions, and support triage. While improving learning over time, this assistant will use an AI architecture based on important agentic patterns to provide real-time, context-aware medical help. This demo will look at design patterns including Reflection, Tool Use, Planning, ReAct (Reasoning and Acting), ReWOO (Reasoning with Open Ontology), and the Multi- Agent Pattern.

How Agentic Architecture and Agentic AI Design Patterns Work Together

Designed to be autonomous with some of the important elements such as Perception, Cognitive, Action, Learning, Collaboration, and Security modules, we considered a modular Agentic Architecture for the healthcare assistant in Demo-1.

This is about learning why the Agentic AI Design Patterns map on these architectural components of Demo-2. Each pattern introduces a degree of reasoning and acting capability that enhance decision-making, versatility, and responsiveness to the environment of the system. Each pattern interacts with the architecture as explained.

1. **Reflection Pattern**

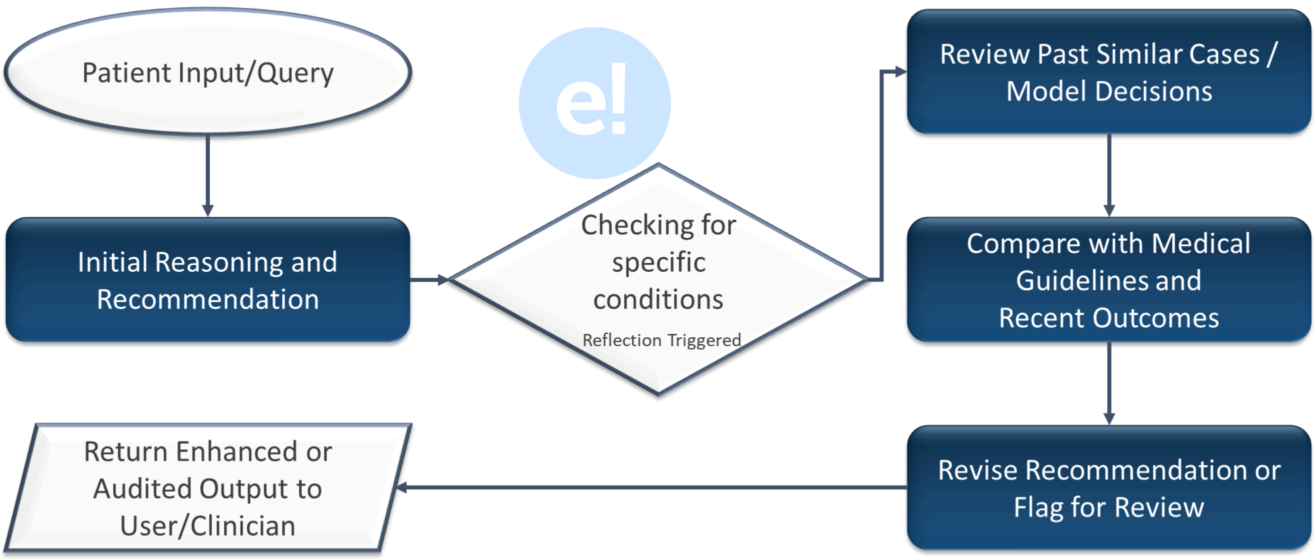
   The Reflection Pattern lets the artificial intelligence grow from past mistakes and improve its decision-making across time.

   Working: Design Pattern + Architecture

- Cognitive Module: integrates Reflection by looking at past patient encounters to determine if specific symptoms or conditions produced normal results, improving future decisions based on what has happened before.
- Learning Module: retains facts from prior cases (like patient responses to treatments or triage recommendations) and helps the system improve its reasoning methods for better future responses.

Example:

Should the healthcare assistant receive multiple complaints about slow reactions to particular symptoms (e.g., "difficulty in breathing"), it examines this data and adjusts her triage protocol to ensure faster prioritizing for forthcoming patients exhibiting similar symptoms.



2. Tool Use Pattern

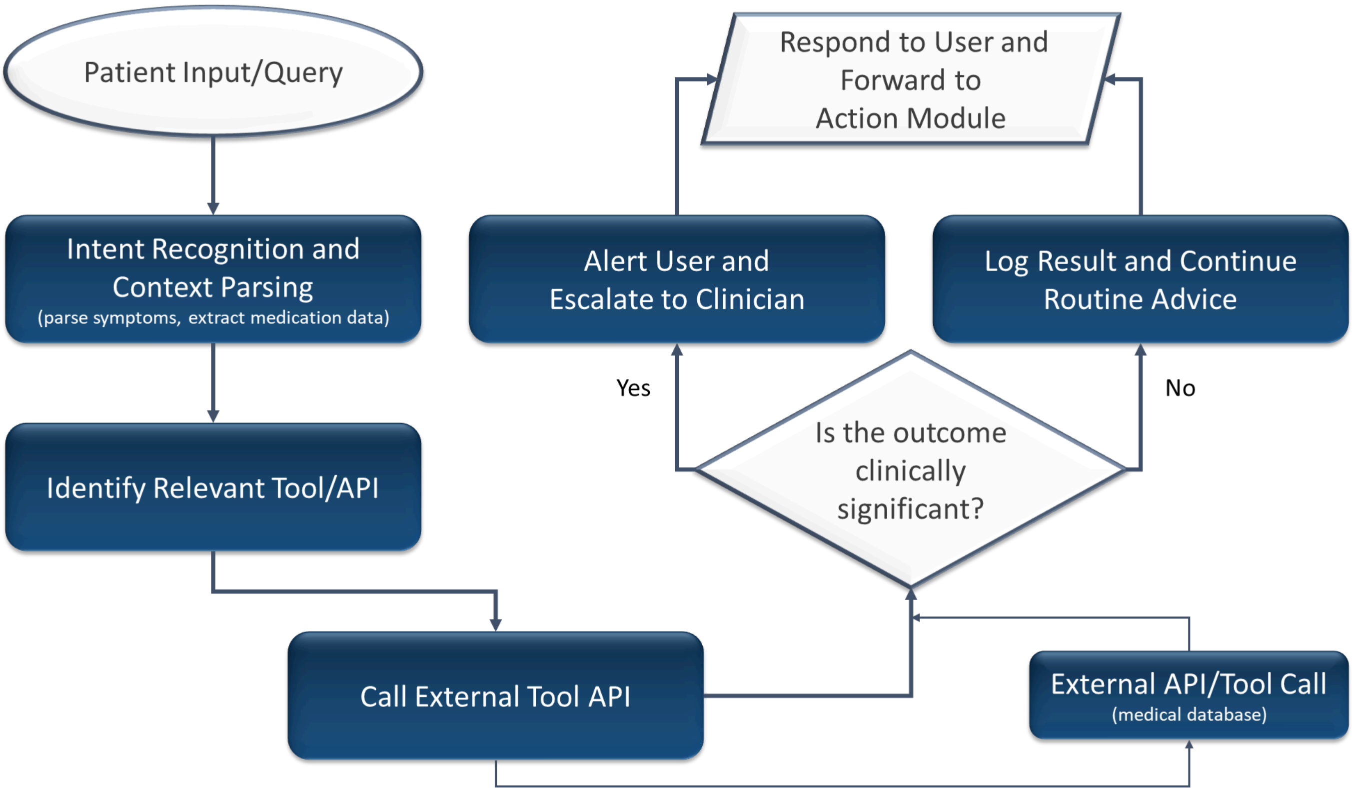
The Tool Use Pattern lets the agent make use of outside tools, databases, or systems to improve its capacity to complete tasks above its natural capacity.

Working: Design Pattern + Architecture

- Action Module: enables Tool Use by connecting the agent to medical decision support systems, diagnostic tools, or electronic health record (EHR) databases to aid in immediate symptom assessment and triage.
- Perception Module: collects initial patient data (e.g., symptoms, medical history) and initiates the Action Module to use external resources (e.g., ICD-10 codes, clinical decision support systems) to improve the diagnosis.

Example:

A patient notes that they are suffering with shortness of breath and chest discomfort. Using the Tool Use Pattern, the assistant searches a medical decision support tool cross-referencing these symptoms with a known database of heart attack symptoms to propose the next actions, such phoning for emergency assistance.



3. Planning Pattern

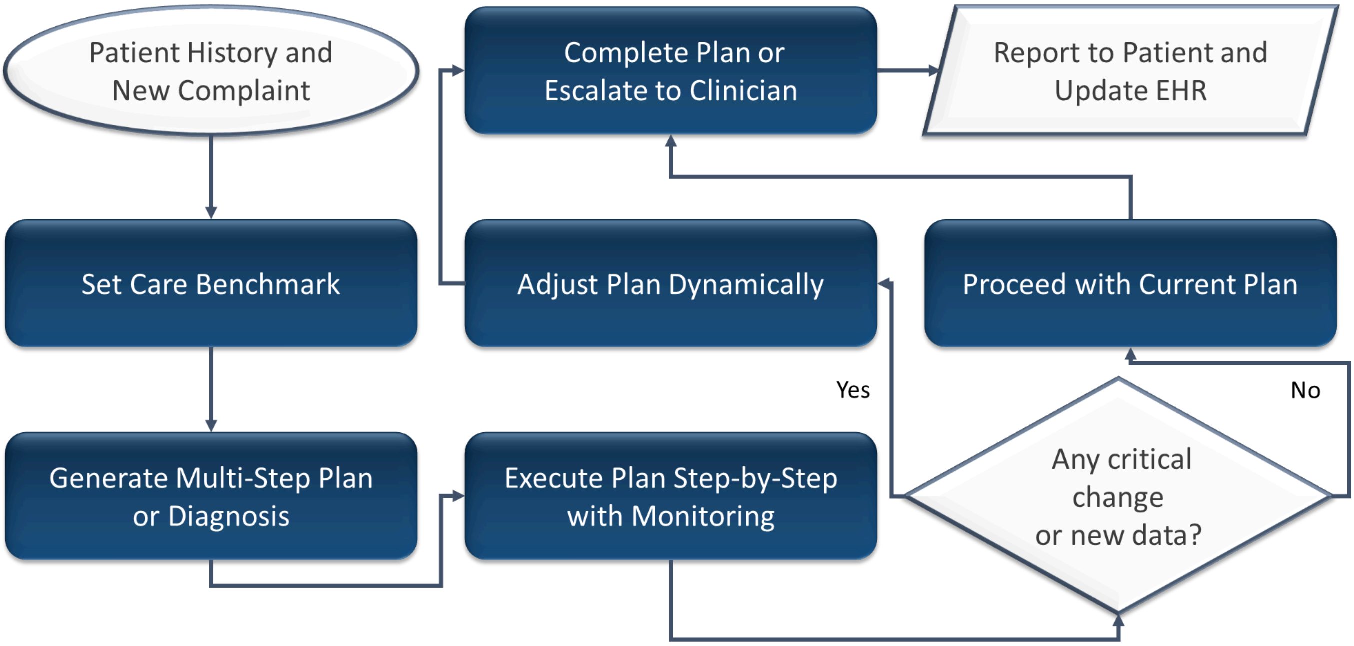
The Planning Pattern helps the AI to create a series of actions required to reach a goal, hence optimizing its decisions depending on context.

Working: Design Pattern + Architecture

- Cognitive Module: assesses the circumstances (e.g., patient symptoms) and devises a strategy for the required actions (e.g., contact an ambulance, inform healthcare providers, request additional information).
- Action Module: carries out the intended actions, coordinating among various systems (e.g., reaching out to emergency services, documenting the interaction in the patient's record).

Example:

Should the artificial intelligence find that a patient's symptoms could be connected to a heart attack, the system arranges the actions to guarantee the patient receives the appropriate treatment: calling emergency services, telling their doctor, and entering the pertinent data into the patient's EHR.



4. ReAct (Reasoning and Acting)

The ReAct pattern lets the agent act and reason concurrently, hence facilitating real-time decision-making. This implies that, depending on its logic, the agent may respond right away as it learns about the condition of a patient. ReAct guarantees that the agent not only thinks but also acts fast to help to reduce hazards or assist others.

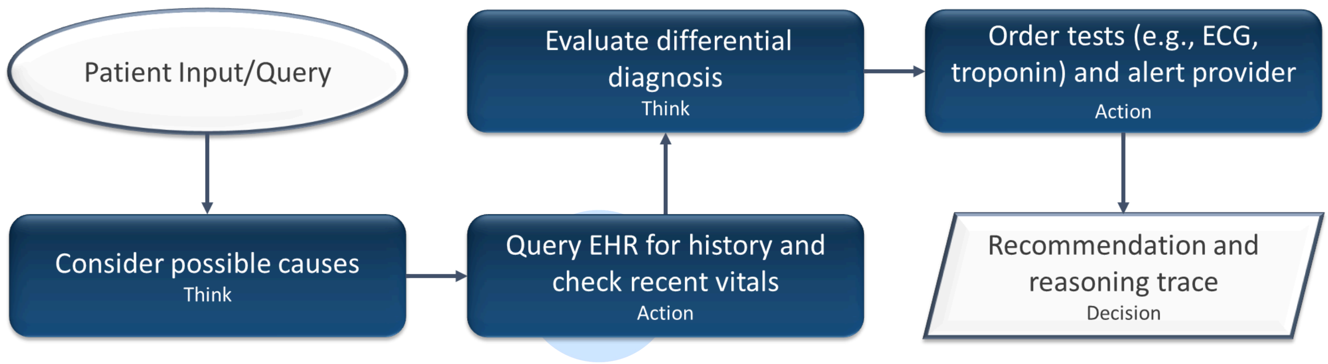
Working: Design Pattern + Architecture

- Cognitive Module: analyzes the patient's symptoms instantly (e.g., chest discomfort, difficulty breathing). It assesses the seriousness of the symptoms and creates a plan of action grounded in available information (for instance, determining whether the patient may be experiencing a heart attack or another urgent issue).
- Action Module: After the reasoning process is complete, this module promptly acts on the insights given by the Cognitive Module. This might entail activating emergency procedures, like contacting

emergency medical services, informing the patient's healthcare provider, or guiding the patient on what to do next.

Example:

Should a patient describe symptoms including shortness of breath and chest discomfort, the system instantly employs ReAct to evaluate the degree of the symptoms. The Cognitive Module evaluates the risk of a life-threatening illness (such as a heart attack) and immediately tells the Action Module to notify emergency personnel, therefore guaranteeing timely action.



5. **ReWOO (Reasoning with Open Ontology)**

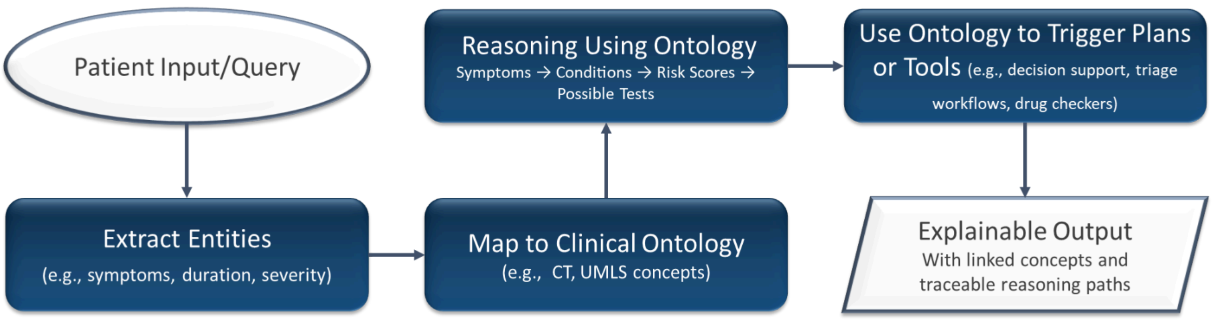
ReWOO presents a more open-ended thinking style that lets the artificial intelligence keep changing its knowledge base as fresh facts or medical studies come in handy. This helps the agent to manage newly developing medical conditions, adapt to changes in medical practice, and over time enhance its decision-making. ReWOO maintains the AI flexible enough to handle novel, unanticipated events unlike conventional systems depending on fixed knowledge.

Working: Design Pattern + Architecture

- Learning Module: is essential in ReWOO, consistently enhancing the system's knowledge base with fresh information, including recent medical discoveries, newly identified conditions, or updates in treatment guidelines. This guarantees that the agent's reasoning stays current and mirrors the most recent healthcare trends.
- Cognitive Module: When analyzing patient symptoms, this module can access the updated knowledge base to facilitate better decision-making, ensuring that the AI avoids forming outdated conclusions. ReWOO enables the system to make inferences based on the latest comprehension of the condition, as it adjusts and gathers insights from the advancing medical knowledge.

Example:

ReWOO helps the Learning Module to include fresh information or case studies on the illness when a patient shows with an uncommon symptom, such as a new or poorly understood side effect from a medication. The Cognitive Module then works through the patient's condition using this revised information to generate a recommendation reflecting the most recent medical understanding.



6. Multi-Agent Pattern

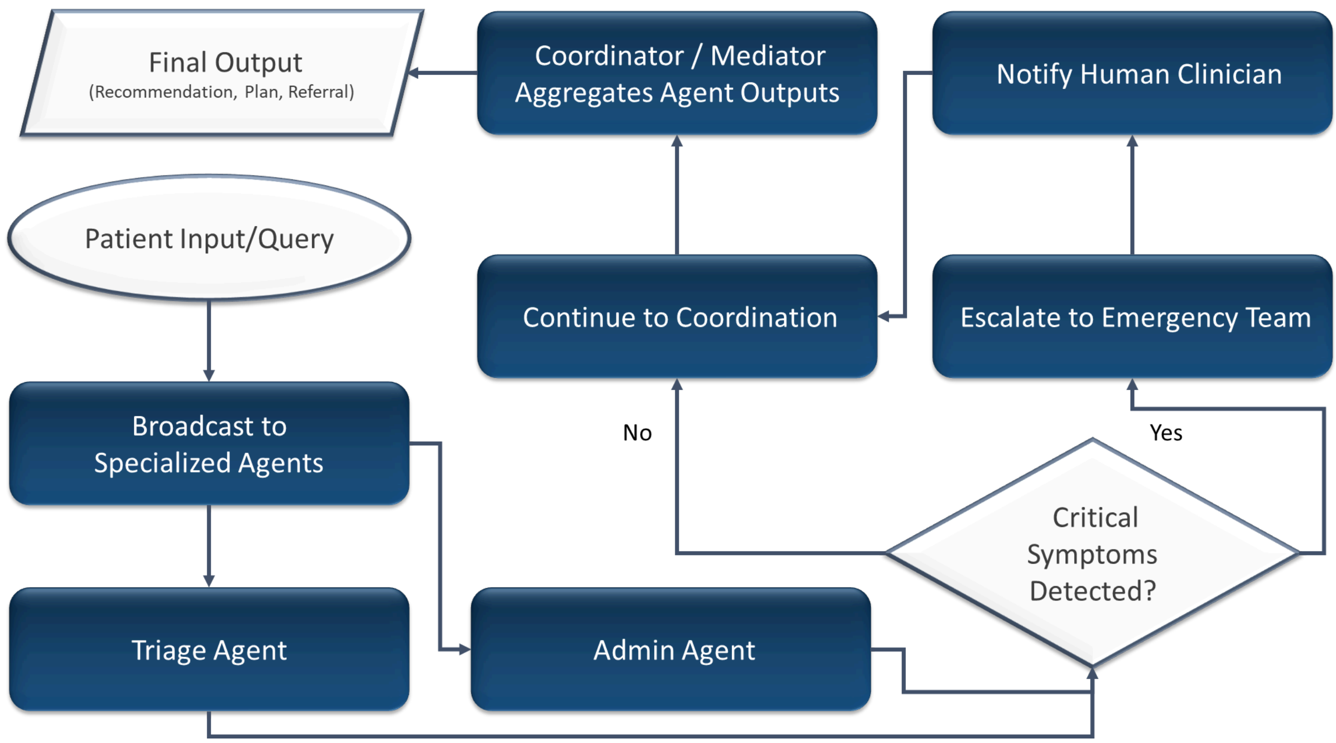
The Multi- Agent Pattern helps several agents to concentrate on particular tasks and cooperate to solve complex problems, hence improving scalability and efficiency.

Working: Design Pattern + Architecture

- Collaboration Module: manages interaction and cooperation between various agents. In this scenario, agents assigned to functions like triage, prescriptions, billing, and follow-up appointments can operate simultaneously.
- Action Module: of each specialized agent manages particular tasks. For instance, one agent could contact emergency services, another might create a prescription, and another may update the patient's record in the database.

Example:

When a patient needs urgent care in a complicated situation, the Multi-Agent System operates as follows: one agent manages triage, another manages emergency service dispersion, and still another real-time electronic health record (EHR) update for the patient guarantees all actions are coordinated and executed quickly.



Design Considerations for the Healthcare System

Implementing these design patterns in a healthcare environment calls for careful consideration of important factors to guarantee the system is scalable, maintainable, and flexible:

1. **Modularity:** In modular architecture, the agentic design patterns fit well since they let any agent or module operate on its own. This modularity helps scaling and maintenance to be simpler.
2. **Scalability:** Without compromising performance, the Multi- Agent Pattern and ReAct design pattern let for a scalable system that can meet rising demand (e.g., processing more patients, more data sources).
3. **Adaptability:** ReWOO and Reflection help the artificial intelligence system to learn from prior encounters and adapt to new situations, therefore always enhancing its reactions and efficiency.

In Demo-2, tried to understand the application of Agentic AI Design Patterns within a healthcare assistant system that utilizes an Agentic Architecture. The combination of Reflection, Tool Use, Planning, ReAct, ReWOO, and the Multi-Agent Pattern enables the system to think, act, adjust, and expand while guaranteeing that the patient gets better and prompt care. This method provides a versatile, adaptable, and scalable solution to the changing demands of healthcare provision.